Lecture 7: Experiments and what random assignment buys

Empirical Methods for Law and Economics | EDLE

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Today's questions

Did the fine increase late arrivals?

How sure can we be with ten day-care centres?

- By the end of today you will be able to:
  - explain what random assignment guarantees and what it does not;
  - compute a difference-in-differences from four averages and from a regression;
  - choose standard errors that count the independent units;
  - judge the validity of an experiment and design one.

DISCUSS Random assignment

- Gneezy and Rustichini (2000) observed 10 private day-care centres in Haifa for 20 weeks. From week 5 to week 16, six centres, "selected randomly", fined parents who came late. The other four centres never had a fine.

Questions:

1. What does the random selection of the six centres guarantee?
2. What does it *not* guarantee, with only ten centres?
3. Why not let the principals decide whether to introduce the fine?

Gneezy, U. and Rustichini, A. (2000), *Journal of Legal Studies* 29(1): 1–17, Section II.B.

1. Which centres got the fine did not depend on anything related to late arrivals, observed or not. Any difference between the two groups, in their levels or in how they would have changed without the fine, is due to chance alone.

2. It does not guarantee that the two groups are similar. With ten centres, chance alone can produce groups that differ, so we compare them in the weeks before the fine.

3. A principal who chooses a fine may run a centre where lateness is already a problem, or already improving. The comparison would then mix the effect of the fine with the reasons for choosing it.

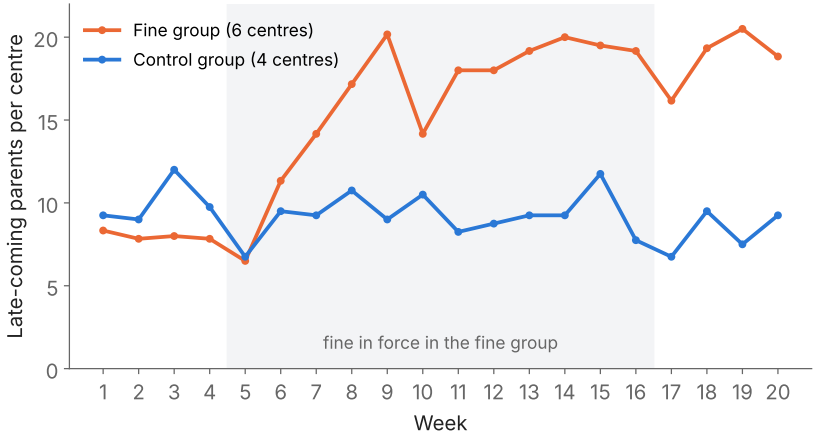
STATA Guided Stata 1: before the fine

```
use "$course_data/daycare_fines.dta", clear
describe
tabstat late children if week <= 4, by(fine_group) statistics(mean n)
preserve
collapse (mean) late, by(fine_group week)
twoway (line late week if fine_group == 1) (line late week if fine_group == 0), ///
    xline(4.5 16.5) legend(order(1 "Fine" 2 "Control"))
restore
```

Weeks 1–4	Fine group	Control group
Late-coming parents per centre and week	8.0	10.0
Children per centre	33.7	33.8

- One row is one centre in one week: $10 \times 20 = 200$ rows, transcribed from Table 1 of the paper.
- Comparing the groups before the treatment is a **balance check**. Same number of children; the fine group starts with two fewer late parents.

Late arrivals, week by week



- Before week 5 both lines are roughly flat, about 2 apart. From week 6 the fine group rises; the control group does not.

PREDICT Four averages

- Weeks 1–4: fine group 8.0, control group 10.0.

Predict, from the figure:

1. the average of each group in weeks 5–16;
2. the difference-in-differences: (change in the fine group) − (change in the control group).

Why not simply compare the two groups in weeks 5–16?

STATA Guided Stata 2: difference-in-differences

```
generate byte during = (period == 2)
tabulate fine_group during if week <= 16, summarize(late) means
regress late i.fine_group##i.during if week <= 16
```

Mean late	Weeks 1–4	Weeks 5–16
Fine	8.00	16.44
Control	10.00	9.23

DiD
$= (16.44 - 8.00) - (9.23 - 10.00) \approx 9.22$
(computed from unrounded means)

regress, 160 rows	Coef.	(SE)
Constant	10.00	(1.41)
1.fine_group	−2.00	(1.82)
1.during	−0.77	(1.62)
1.fine_group#1.during	9.22	(2.10)

- `##` adds the two indicators and their product. The coefficient on the product is the difference-in-differences.

Reading the regression

$$\text{late}_{it} = \beta_0 + \beta_1 \,\text{Fine}_i + \beta_2 \,\text{During}_t + \beta_3 \,(\text{Fine}_i \times \text{During}_t) + u_{it}$$

- $\beta_0 = 10.00$: the control group before the fine.
- $\beta_1 = -2.00$: the fine group minus the control group, before the fine.
- $\beta_2 = -0.77$: the change in the control group.
- $\beta_3 = 9.22$: the change in the fine group beyond that of the control group. With the fine, about 9 more late-coming parents per centre and week, from a starting level of 8.
- Comparing the groups in weeks 5–16 only gives $16.44 - 9.23 \approx 7.22$: it ignores that the fine group started 2 lower.
- Assumption (**parallel trends**): without the fine, the fine group would have changed like the control group. The weeks before the fine are our evidence.

DISCUSS 160 rows, standard error 2.10

Questions:

1. The regression uses 160 rows. How many independent units are there?
2. Which assumption of Lecture 6 is in doubt?
3. What would a referee ask the authors to do?

ANSWERS 160 rows, standard error 2.10

1. Ten. The fine was assigned to centres, not to weeks, and the 16 weeks of one centre share the same parents, children and staff.

2. Assumption 2, independent observations. The usual standard error treats the 160 rows as 160 independent pieces of evidence.

3. To compute standard errors that count centres: standard errors **clustered** by centre, or a comparison of one number per centre.

STATA Guided Stata 3: counting centres

```
regress late i.fine_group##i.during if week <= 16, vce(cluster centre)
preserve
generate late_before = late if week <= 4
generate late_fine = late if week >= 5 & week <= 16
collapse (mean) late_before late_fine, by(centre fine_group)
generate change = late_fine - late_before
list centre fine_group late_before late_fine change
ttest change, by(fine_group)
restore
```

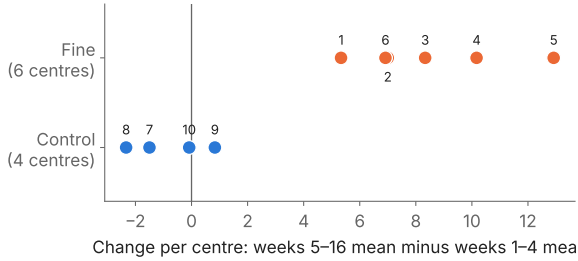
Difference-in-differences: 9.22	SE	t	df	p
Usual SE, 160 rows	2.10	4.39	156	< 0.001
Clustered by centre	1.26	7.30	9	< 0.001
One change per centre, `ttest` (6 vs 4)	1.50	6.15	8	< 0.001

- `ttest` computes control minus fine, so Stata prints -9.22 and $t = -6.15$.

Why the standard error went *down*

- Before the fine, centres ranged from 5 to 13 late parents per week. The usual formula treats each centre-week as an independent draw, with a spread of about 5.6 late parents, and part of that spread is the permanent difference between centres.
- The difference-in-differences compares each centre with itself, so these permanent differences cancel. The clustered standard error and the `ttest` use only how much the change differs across centres of the same group: a standard deviation of about 2.3.
- Clustering counts the right units. It can make standard errors larger or smaller.
- With ten centres, the clustered standard error is itself imprecise. The `ttest` on ten changes is the most transparent check, and it gives the same conclusion.

A test that uses only the random assignment



- All six fined centres increased, by 5.3 to 12.9; the four control centres changed by -2.3 to $+0.8$.
- Suppose the fine had no effect. Then the six fined centres are just one of the 210 ways to choose six centres out of ten, all equally likely. Only one of the 210 puts the six largest increases in the fine group: probability $1/210 = 0.005$ (one-sided; counting also the opposite extreme, $2/210 = 0.01$).
- This is a **randomization test**. It needs no normal distribution and no large sample, only random assignment.

DISCUSS After the fine was removed

- Weeks 17–20, without any fine: fine group 18.7, control group 8.3 (weeks 1–4: 8.0 and 10.0). Difference-in-differences: 12.5.

Questions:

1. Late arrivals did not fall back when the fine was removed. Give two explanations.
2. Can these data tell the two explanations apart?
3. What does this imply for a legislator who wants to "try a fine for a while"?

ANSWERS After the fine was removed

1. **Information**: before the fine, parents did not know what could happen if they came late, perhaps something serious. The fine told them that the worst consequence was a small payment, and removing the fine does not erase that information (Section IV.A). **Social norms**: the fine turned a favour to the teachers into a service with a price, and the price stays in parents' minds: "once a commodity, always a commodity" (Section IV.D).

2. No. Both explanations predict the same pattern. Telling them apart needs another design, for example one that gives some parents information about the cost of lateness without a fine.

3. A temporary sanction can change behaviour permanently. Removing it may not bring back the situation before it was introduced.

DISCUSS Internal and external validity: the referee's view

(a) The study began with 12 centres; two were dropped because their recordings were incomplete (footnote 7).

(b) Late arrivals were registered in the centres, where registering late parents was common practice; teachers knew about the fine, not about the study.

(c) The fine was NIS 10 per child for a delay of 10 minutes or more; a baby-sitter earned NIS 15–20 per hour.

(d) The fine was announced as a decision of the centre (Appendix A). When it was removed, parents who asked were told it had been a trial; no parent was asked for consent.

For each point: does it threaten internal validity, external validity, or is it an ethical issue? Which one would you raise as a main comment?

Gneezy, U. and Rustichini, A. (2000), Section II.B, footnote 7 and Appendix A.

ANSWERS Internal and external validity: the referee's view

(a) **Internal**, and the main comment. If the two centres were dropped after the fine started, and the fine affected their recording, the remaining groups are no longer randomly assigned. Ask in which group they were, when they were dropped, and whether the six fined centres were drawn among the 12 or among the remaining 10.

(b) **Internal**. Once lateness had a price, staff in fined centres may have registered it more carefully. Registering was already common practice, but a referee can ask whether it changed with the fine.

(c) **External**. The fine was small. A larger fine might deter: the study shows that a fine *can* increase lateness, not that fines do so in general.

(d) **Ethics**. Parents did not consent. Today such a study needs ethical approval and a reason for not informing participants.

DISCUSS Design an experiment

- A court wants fewer defendants to miss their hearings. Proposal: a text-message reminder a few days before the hearing.

Design the evaluation:

1. What do you randomise: defendants or courts? How many independent units does each choice give?
2. What is the outcome, and who records it?
3. What would you check before looking at the results?
4. Is it ethical to send reminders to only half of the defendants?

ANSWERS Design an experiment

1. Defendants, within the same courts: as many independent units as defendants, thousands in a busy court. Randomising courts gives as many units as courts, a handful, however many defendants each court has.

2. Whether the defendant appears, taken from court records by staff who do not know who received a message.

3. Balance: compare the two groups on what was recorded before the draw (age, past missed hearings, type of charge). The differences should be small. Write down the analysis before seeing the results.

4. The control group gets what defendants get today. While the effect of reminders is unknown, this is defensible; once it is known, everyone should receive them.

Laboratory

- Worksheet: `exercises/07_experiments.do`. Data: `daycare_fines.dta`.
- Tasks:
  1. the difference-in-differences after the fine was removed;
  2. all three periods in one regression;
  3. a placebo test: pretend the fine started in week 3 and estimate the difference-in-differences on weeks 1–4; it should be close to zero;
  4. late arrivals per 100 children;
  5. the randomization test by hand;
  6. one referee comment and one experimental design.

Exit check

1. What does random assignment guarantee, and what does it not guarantee with ten units?
2. Write the difference-in-differences of the fine (9.22) as a sentence.
3. Why can clustered standard errors be smaller than the usual ones?

Next lecture

- Lecture 8 (Friday): legal texts as data. How to turn constitutions into a measure, and how to check that the measure means what we think.
- Read Gentzkow, Kelly and Taddy (2019): Sections 1 and 2; Section 3.5.2.
- Bring your draft comments: we revise them in class.